VI) Common Types of Cyberattacks

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- Spear-phishing
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- Eavesdropping
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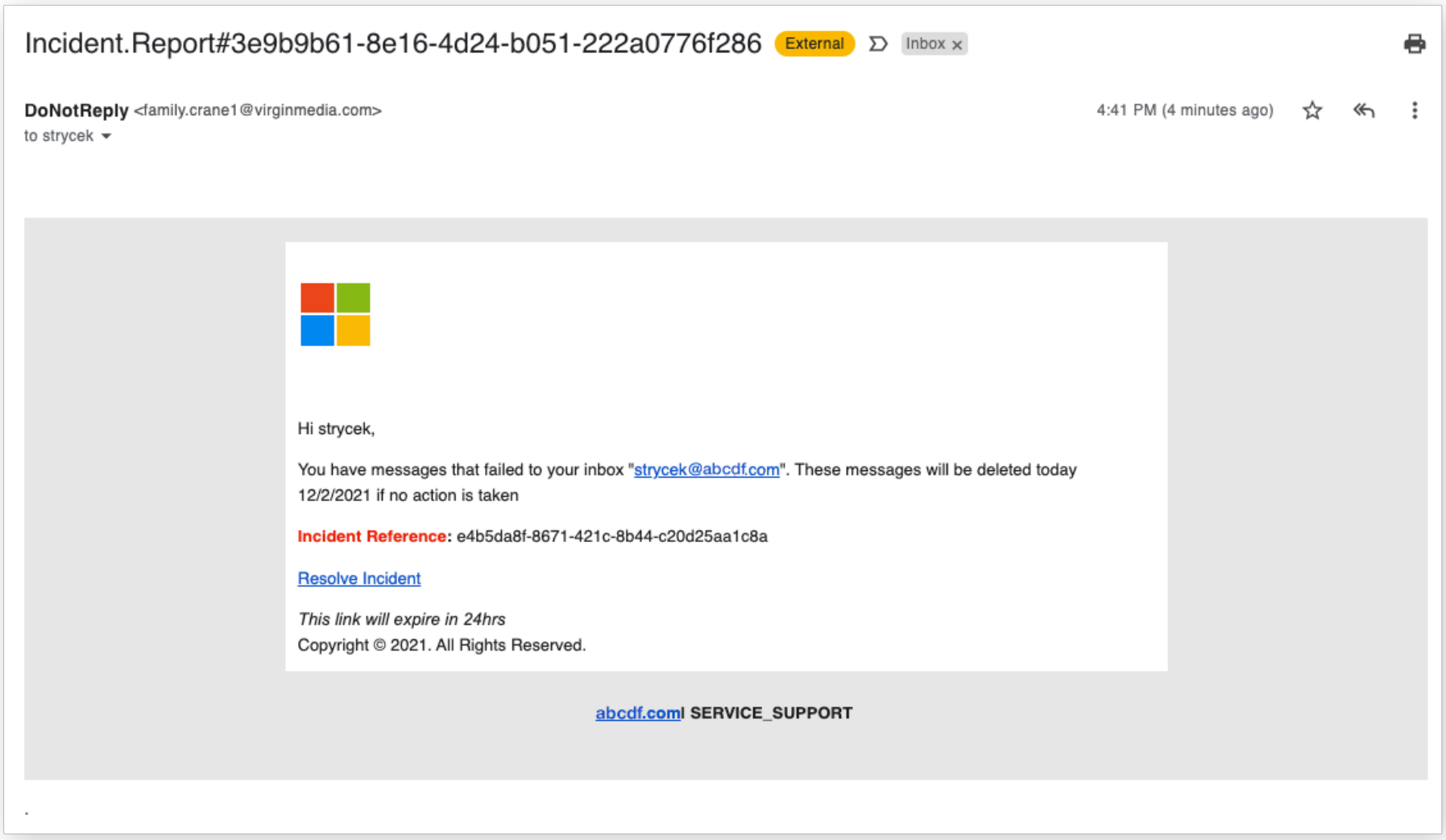
Spear-phishing

What is it? An attempt to **steal sensitive data**, such as account or financial information, from targeted victims.

How it works: An attack requires significant research, which often involves acquiring personal information about the victim. This is typically done by accessing social media accounts to discover information such as their name and email address, who their friends are, their hometown, their employer,ect... Attackers then disguise themselves as someone their victim trusts, usually a friend or colleague, and attempt to acquire sensitive information via email or instant messaging tools

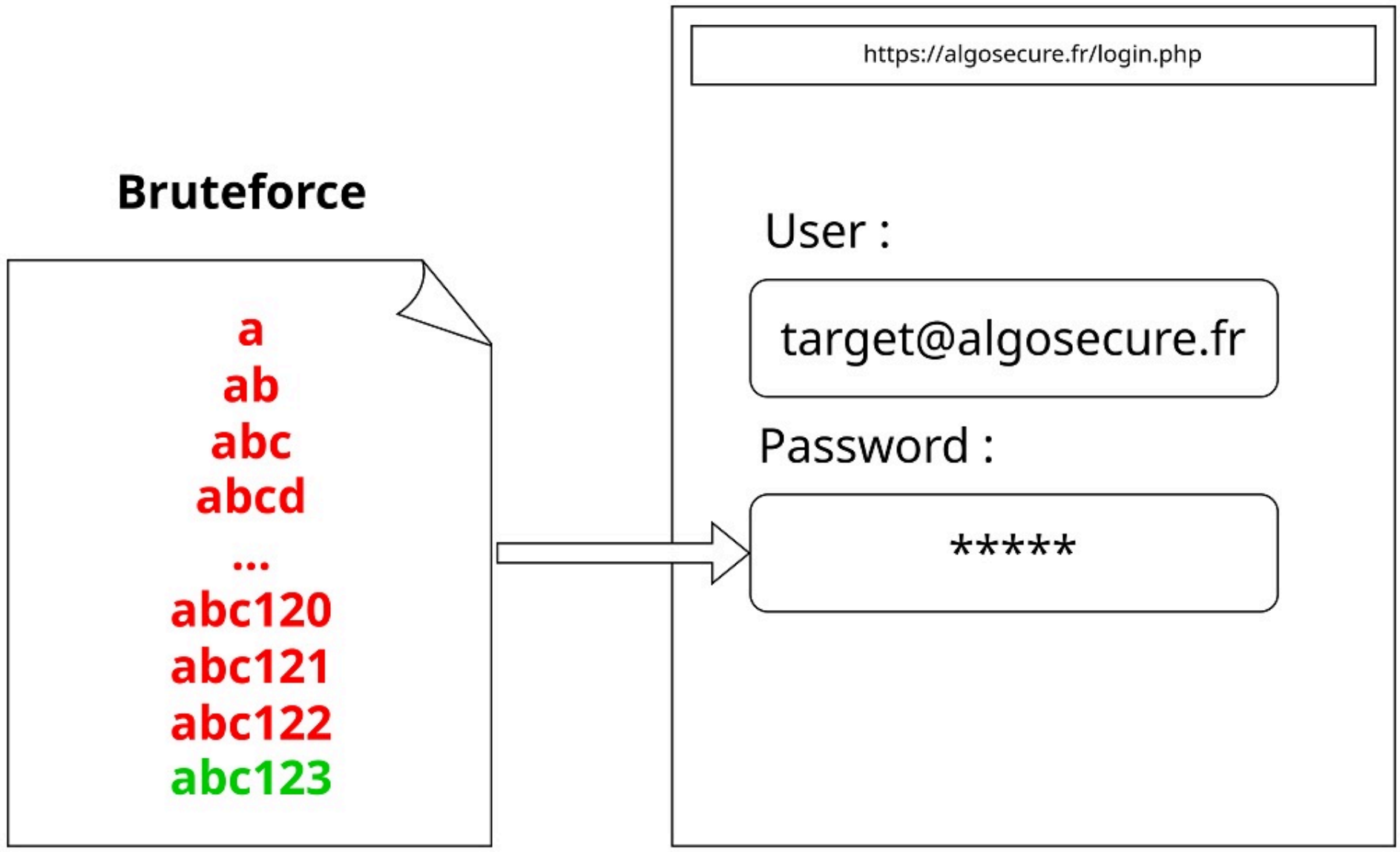
Example: A fake email want you to open a malicious link .



How to protect yourself:

- Be vigilant about emails received : (Sender addresses ; unusual requests for information ; spelling mistakes)
- Verify sender identity before replying.
- Equip yourself with email protection solutions.

Brute Force



What is it? A hacking method that uses trial and error to crack passwords, password credentials, and encryption keys. It is a simple but reliable tactic for gaining unauthorized access to individual accounts and organizational systems and networks.

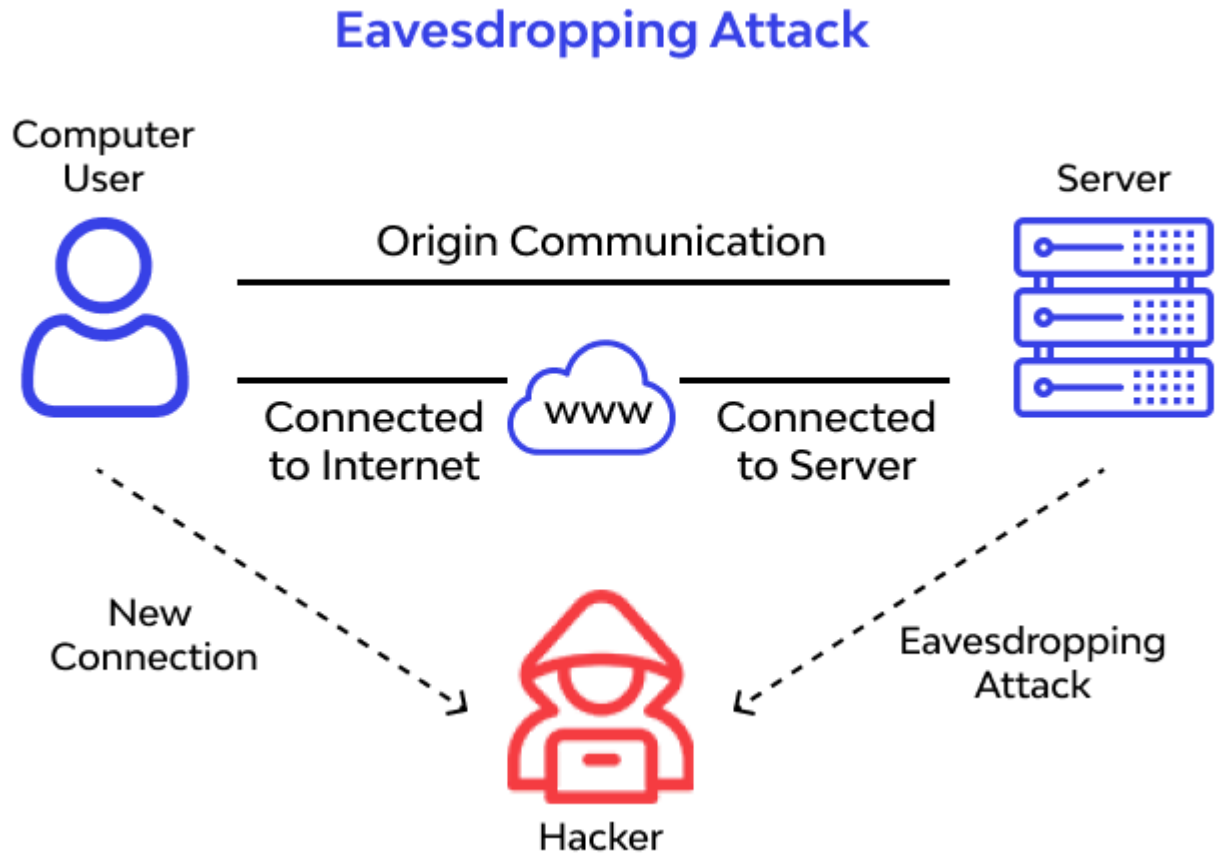
How it works: The hacker tries multiple usernames and passwords, often using a computer to test a wide range of combinations, until they find the correct login information.

Example: Attempting every possible code to unlock an account.

How to protect yourself:

- Create strong, multi-character passwords.
- Avoid common passwords.
- Use Word Managers.

Eavesdropping



What is it? Attacks that typically involve the use of various devices to listen conversations and review network activity.

How it works: Attackers can use devices that pick up sound or images, such as microphones and video cameras, and convert them into an electrical format to eavesdrop on targets. Ideally, it will be an electrical device that uses power sources in the target room, which eliminates the need for the attacker to access the room to recharge the device or replace its batteries.

Example: A good example of the impact eavesdropping attacks can have is the increasing use of digital assistants like Amazon Alexa and Google Home. These assistants make users' lives easier but are also easy for attackers to eavesdrop on and gain private information.

How to protect yourself:

- Be sure that employees/peoples are aware of the risks and dangers.
- Avoid public Wi-Fi for sensitive operations.
- Update your software

Ransomware

What is it? It's a malware designed to deny a user or organization access to files on their computer. By encrypting these files and demanding a ransom payment for the decryption key.

How it works:

1. Infection (Propagation)

Ransomware enters a system via:

- Phishing emails (malicious links or attachments).
- RDP (remote access via stolen credentials).
- Security vulnerabilities, such as EternalBlue, exploited by WannaCry.

2. Data Encryption

Once the system is infected, the ransomware:

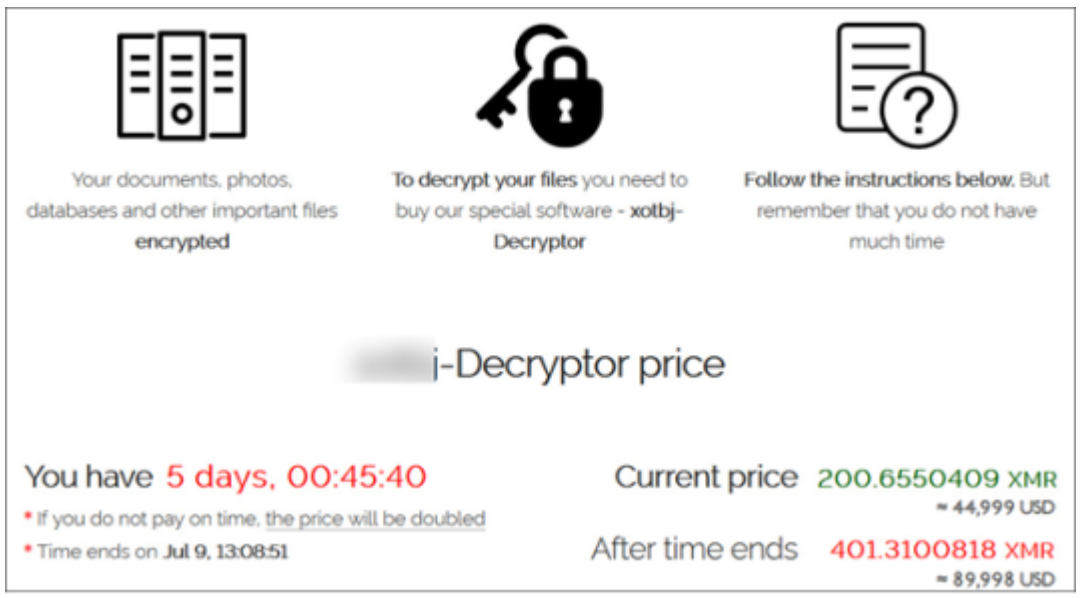
- Encrypts files with a key controlled by the attacker.
- Sometimes deletes backups and hidden copies to prevent recovery.
- Targets certain files to avoid rendering the system unusable.

3. Ransom Demand

After encryption:

- A message is displayed (modified wallpaper or text file).
- The user is asked to pay in cryptocurrency.
- If paid, the attacker may provide a decryption key or tool.

Example:



How to protect yourself:

- Do not click on suspicious links or attachments.
- Use antivirus/antimalware protection.
- Regularly back up important files.

Contact

- Email: hugo.hbt@hotmail.fr
- Phone: 06.XX.XX.XX.XX
- Location: Noisy-le-Grand

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